

RESULTS — for the plane writing session (living; final version stamped before the 08-30 blackout)

Read with: METHODS_draft_2026-08-28.md (UPDATE 08-29 blocks),

PREREG_final_round_robin_2026-08-23.md A9-A19, ADVERSARIAL_AUDIT_2026-08-29.md (what a reviewer will say), WM_CANDIDATES_2026-08-29.md. Statistics: analysis/stats.py (seed is the unit; Welch + exact permutation; min attainable p quoted). Readouts: **sel** = 15 selection ICs (ceiling 14/15, training ICs) — selection only, never a headline; **hold** = 15 ICs ($\subset$ training ICs, A8); **rnd** = 30 random ICs. Headline = selected checkpoint on hold+rnd (PREREG §5/A7). Worlds: **old** = matched_v2 (RLPD, r2dreamer, dv3 to date); **corrected** = matched_w3 gc_kp4_riser3_shelf6 (DP n=10; r2dreamer gate s80-83). One world per table (A19). Version stamp at the bottom.

1. DP (Ierobot Diffusion Policy, state-only) — HEALTHY, the one settled learner

```
world arm n hold rnd
-----
corrected (seeds 20-29) dH 10 0.887 0.547
corrected dDP 10 0.873 0.487
```

- (1a) at matched N=56: hold flat (+0.013); rnd human edge +0.06, Welch CI [+0.006, +0.114], exact-perm p 0.041 — ONE unadjusted comparison, top-up not pre-registered as this test (AUDIT_results §4). Old world n=5 replicate: no spread.
- (1b) not applicable to DP by design (no IL on failures, PAPER_PLAN §3). (2) not applicable (no reward).
- 35 further DP jobs are user-held (replicates); not needed for the claim above.

2. RLPD (SB3 SAC subclass, E=10 Z=2 UTD 10, 50/50 demo batches, LayerNorm critics)

Recipe restart (A17): every $\gamma = 0.998$ run is diagnostic-only — 69/74 diverged (critic loss 10^2 – 10^5). At the published $\gamma = 0.99$ the learner is healthy on human demos. All numbers below are $\gamma = 0.99$, old world, sparse unless stated, seeds ≥ 30 .

```
arm seed max critic loss watchdog trips selected: hold / rnd final: hold / rnd
-----
dH 30 0.018 0 14/15 / 20/30 (12 h limit hit during final eval)
dH 31 0.016 0 14/15 / 21/30 14/15 / 21/30
dH 32 0.027 0 14/15 / 19/30 11/15 / 19/30
dDP 30 22.1 4 10/15 / 14/30 10/15 / 14/30
dDP 31 83.6 5 13/15 / 12/30 13/15 / 12/30
dDP 32 0.043 0 14/15 / 20/30 10/15 / -
dH (UTD 5) 30-32  $\leq 0.045$  0 14/15 / 20, 26, 21 final hold 14, 8, 14
dDP (UTD 5) 30-32 530 / 65 / 0.037 8 / 6 / 0 8/15 14/30; 8/15 7/30; 13/15 14/30
y 0.998 UTD 5 (control) dH 30,31; dDP 30 292-1.5e5 6-9 hold 0-2/15
```

- Reading so far (n=3, NOT a claim): human demos → stable critic 3/3; DP demos → 2/3 diverge with partial performance. Demo sets are indistinguishable on every tape statistic (tape_census_2026-08-29.txt), so this is not a format artefact.
- **Pre-registered statistic (A16):** divergence rate (Fisher) + LAST-checkpoint rnd-30 (Welch/perm). Predictions: dH 0-1/8 vs dDP $\geq 3/8$; LAST rnd dH > dDP by ≥ 0.10 . Matrix in flight: dH/dDP sparse s33-37, dHHfails/dDPfails s30-37, dH/dDP dense s30-35, dHsucc_dup/dDPsucc_dup s30-33 (row-matched controls). First s33-37 headlines: see §2.1 (updated at each check). Early: dH s33 2 mild trips (CL 0.16), s36 CL 2.9 (crosses the A16 threshold) — dH is not immune.
- Watchdog trip count is NOT valid on dense arms (shaping lifts Q); use max critic loss + final≈selected there.

2.1 g99 matrix readouts (appended as they land; sparse, old world)

```
arm seed max CL trips selected hold / rnd final hold / rnd
-----
dH 30 0.018 0 14/15 / 20/30 (12 h limit hit during final eval)
dH 31 0.016 0 14/15 / 21/30 14/15 / 21/30
dH 32 0.027 0 14/15 / 19/30 11/15 / 19/30
dH 33 0.157 2 11/15 / 19/30 14/15 / 21/30
dH 34 0.138 2 15/15 / 17/30 15/15 / 17/30
dH 35 0.017 0 14/15 / 20/30 14/15 / 20/30
dH 36 4.42 4 14/15 / 25/30 14/15 / 25/30
dH 37 0.019 1 14/15 / 24/30 14/15 / 24/30
dDP 30 22.1 4 10/15 / 14/30 10/15 / 14/30
dDP 31 83.6 5 13/15 / 12/30 13/15 / 12/30
dDP 32 0.043 0 14/15 / 20/30 10/15 / -
dDP 33 39.2 6 12/15 / 17/30 12/15 / 17/30
dDP 34 0.03 0 12/15 / 16/30 12/15 / 16/30
dDP 35 5.28 4 12/15 / 13/30 12/15 / 13/30
dDP 36 18.5 7 11/15 / 17/30 11/15 / 17/30
dDP 37 4.56 4 13/15 / 18/30 13/15 / 15/30
```

HEADLINE (1a) for RLPD — pre-registered A16, old world, sparse, γ 0.99, seeds 30-37 (COMPLETE n=8 v 8; dH s30 has no final-ckpt readout, dDP s32 none → LAST n=7 v 7):

```
statistic dH dDP diff 95 % CI Welch p exact perm p
-----
selected ckpt, rnd-30 0.688 (n=8) 0.529 (n=8) +0.158 [+0.062, +0.254] 0.003 0.004
LAST ckpt, rnd-30 (A16 statistic) 0.700 (n=7) 0.495 (n=7) +0.205 [+0.111, +0.299] 0.001 0.002
selected ckpt, hold-15 0.917 (n=8) 0.808 (n=8) +0.108 [+0.022, +0.195] 0.018 0.032
divergence (max critic loss  $\geq 1$ ) 1/8 6/8 - - Fisher 0.041
```

A16 predictions ("dH 0-1/8 vs dDP $\geq 3/8$ "; "LAST rnd dH > dDP by ≥ 0.10 ") both met. Relative loss on random ICs 25-29 %. Tape census rules out format artefacts (§2 of METHODS UPDATE); the remaining mechanism candidate is demo-state coverage (DP-teacher trajectories are narrower in state space than human ones — the same argument as the exposure/coverage story). Per-seed: dH selected hold 14,14,14,11,15,14,14,14 / rnd 20,21,19,19,17,20,25,24; dDP hold 10,13,14,12,12,12,11,13 / rnd 14,12,20,17,16,13,17,18. LAST rnd where a final readout exists: dH 21,19,21,17,20,25,24; dDP 14,12,17,16,13,17,15. One dH seed (s36) crossed the critic-loss threshold yet scored 14/15, 25/30 — a divergence event is not a performance failure; the performance statistic is the one to lead with, the divergence rate is the mechanism. Row-matched control dHsucc_dup (56 successes + 8 longest own successes duplicated), sparse, first 3 seeds (01:50 08-

30): s30 hold 13 rnd 17, s31 15/23, s32 14/23 — all clean (max CL ≤ 0.08); i.e. adding $\sim 1.7k$ rows of duplicated human successes leaves dH at its level (rnd 0.70 vs 0.69). Still to come (during the blackout): dHsucc_dup s33, dDPsucc_dup s30-33, dense arms (question 2), fails arms (1b).

3. r2dreamer (in-house DreamerV3; replay critic loss = official DV3 v2 component; return clamp = deviation)

- **Old world, dense, 3M (historic block, seeds 50-53, 100-103, 80-87 dR2D):** selected checkpoints reach 13-15/15 hold on dH and dR2D; **sparse never learns** (0/7,700 episodes). LAST-checkpoint hold re-scores: dH s50-53 = 0, 15, 0, 7; dH s101/103 = 0, 0; dDP s100/102/103 = 1, 10, 14; dR2D s80-83/86/87 = 15, 0, 13, 0, 13, 11; dR2DDPfails = 5, 5, 0, 1, 15, 4 $\rightarrow$ final policies are **bistable** (0 or ≥ 10 of 15 in 16/21); BEST-of-K on sel does the work. Disclose.
- **Corrected world gate (dH/dDP s80-83, dense, 3M, packed, 08-29): IGNITES.** In-job sel max per seed: dH 0.40, 0.93, 0.93, 0.93 (nonzero ckpts 3/17/13/14 of 28); dDP 0.00, 0.87, 0.13, 0.93 (0/17/4/8). Same adjacent-checkpoint bistability. Protocol re-scores (BEST hold/rnd, LAST hold; n12_rescore/_w3_) $\rightarrow$ §3.1.
- **Standard arm (clamp off, repval kept) NOCLAMP dH/dDP s122/123 — RUNS AWAY.** At $\sim 0.7M$ sim steps train/val = 825, 343 (dH), 226, 115 (dDP) against a 100 maximum — the same critic runaway as dv3, now in r2dreamer with the official replay critic loss present. So the clamp is what makes r2dreamer learn; neither torch port is stable under the published recipe on this task. Write-up: the WM arm = DreamerV3-with-return-clamp, disclosed; the standard-recipe negatives (dv3-std $\times 3$, r2d-NOCLAMP $\times 4$) are reported as such. Final numbers land in the +44/+62 h harvests.
- N15 (dR2D + DP-fails, mixed source): direction-only (n 4 v 3, CI spans 0, min p 0.057); mechanism cell, not a matrix cell.

3.1 W3 protocol re-scores (corrected world, dense, 3M; BEST = highest in-job sel among archived ckpts)

```
arm seed sel max BEST hold BEST rnd LAST hold
-----
dH 80 0.40 10/15 20/30 4/15
dH 81 0.93 15/15 25/30 13/15
dH 82 0.93 11/15 22/30 15/15
dH 83 0.93 10/15 16/30 0/15
dDP 80 0.00 1/15 1/30 1/15
dDP 81 0.87 2/15 2/30 5/15
dDP 82 0.13 2/15 3/30 1/15
dDP 83 0.93 15/15 28/30 13/15
```

Seed-level (analysis/stats.py, n=4 v 4): BEST hold dH 0.767 vs dDP 0.333, diff +0.43, CI $[-0.24, +1.11]$, perm p 0.143 (min attainable 0.029); BEST rnd 0.692 vs 0.283, diff +0.41, CI $[-0.26, +1.07]$, p 0.143; LAST hold 0.533 vs 0.333, p 0.63. Reading: in the corrected world r2dreamer learns from human demos on 4/4 seeds (hold $\geq 10/15$) and from DP demos on 1/4; dDP s81's sel 0.87 collapsed to 2/15 on hold (sel-selected checkpoint was a fluke — another reason sel is never a

headline). Direction: human > DP for the world model; NOT significant at n=4 (min p 0.029). Seeds 84-87 per source submitted 08-29 13:30 (finish during the blackout) → n=8 gives min p 0.0002 and ~80 % power at the observed gap.

4. dv3 (dreamerv3-torch, NM512 port) — mechanism found; standard arm under test

- **Failure mechanism:** on every unclamped x100-terminal run `value_mean` runs past the attainable return (263-390 vs 100 by 300k, 3/3 5M baselines, zero success); touchgoal-from-scratch s0 learned to 0.82-0.96 (230-440k) then collapsed as value crossed 100 (132, 140). Critic runaway via bootstrapped reward-head leak; the port lacks the official replay critic loss / EMA regulariser (`WM_CANDIDATES` §1). Old runs' "transients" (N4) are this phenomenon.
- **Standard arm genesis_dv3std** (reward_scale 1 symlog/two-hot, no clamp, train_ratio 512, TL 400, demos terminal 1): s20 first picks 0.22 @284k (0.06, then 0 × 8 samples to 550k), value bounded ~33; s21 0.04 @191k, s22 0.02 @177k, values 30-36. clamp s20: 0.25 @~290k then 0, value pinned 98; fix s20: 0.18 max, sporadic 0.02, value 97. → transient picks under every recipe, none sustained yet (r2dreamer's phenotype before its 1-2M ignition). 1M verdict → §4.1; 3M std pack (s23-25) with resume chain runs through the blackout.
- Disclosures: dv3 jobs before 08-28 21:30 ran without overlays (launcher bug); 5M baselines unaffected.

4.1 dv3 std @1M (appended)

18:30: std s20 @890k — no success since 303k, value now DECREASING (30 → 19); s21 @505k 9 nonzero samples (max 0.22), value 8; s22 @510k 1 nonzero, value 8. Verdict at 1M for s20: NEGATIVE (transient picks, no learning; value 12 and falling at 950k). The launcher's post-training eval failed on s20 (stale timestamp dir from a cancelled job; fixed in the launcher 22:00, but s21 inherits the bug — its protocol eval must be run by hand after the blackout; s22 is clean). 3M pack (s23-25) started 18:00 and runs through the blackout with its resume chain. DEMO3/MoDem cluster smoke attempt launched 18:40 (A18 trigger). 15:50: std s20 @780k (4 nonzero, none since 303k, value 30); s21 @410k 8 nonzero incl. 0.22 @~350k and 0.12 @366k then 0, value 21; s22 @418k 1 nonzero; clamp s20 @847k 7 nonzero, value 97; fix s20 @572k 10 nonzero, value 99. Still no sustained learning; s21 is the most active. Earlier (13:20): std s20 @670k — 4 nonzero success samples total (all ≤ 0.22, none after 303k), value 28 bounded; s21 @310k 5 nonzero (≤ 0.04); s22 @310k 1; clamp s20 @710k 6 nonzero (≤ 0.25), value 97; fix s20 @476k 10 nonzero (≤ 0.18), value 98. No sustained learning under any dv3 recipe yet.

5. Fallback WM (A18)

DEMO3 codebase (TD-MPC2 backbone; MoDem/TD-MPC2 by flag; MIT) prepared locally: converter round-trip verified on dH, env wrapper import-checked;

policy_pretraining=false flag set is actor-BC-free. ~4-6 h to a cluster smoke.
Triggered 08-29 (both standard arms negative): venv installed on the cluster (torch 2.7.0+cu126 kept; torchrl 0.8.1, tensordict 0.8.3), smoke job 3034689 pending → result in `paper/DEM03_SMOKE_2026-08-29.md` / the harvest files.

6. What can be written now (claims with evidence) — see

ADVERSARIAL_AUDIT §7

1. Methods/provenance contribution (one recorder, tape contract, registry, six silent-default bugs caught, run-identity rule).
2. DP: DP-teacher demos $\approx$ human at matched N; small human edge on random ICs (p 0.041, unadjusted).
3. **RLPD (1a): under the published recipe, human demos beat DP-teacher demos on random ICs by +0.21 (LAST ckpt, p 0.001, n 7 v 6; pre-registered A16) and diverge less (1/8 vs 6/8, Fisher p 0.041); recipe (γ) sensitivity dominates everything at γ 0.998.**
4. WM: r2dreamer (return-clamped DreamerV3) learns pick with dense reward + demos in BOTH worlds, checkpoint-bistable, human > DP in the corrected world (0.77 vs 0.33 hold, n=4, p 0.14; n=8 pending); sparse never ignites. Under the published recipe (no clamp) BOTH torch ports run away (dv3-std $\times 3$ no learning; r2d-NOCLAMP values 115-825 vs max 100) — a characterised negative, not a missing experiment.
5. Fails-tape effect: confounded (share $\times 2.7$, length $\times 12$) until the row-matched controls read out.

Nothing yet on (1a)/(1b) for RLPD or WMs beyond the above; nothing on (2) beyond "sparse WM never ignites".

7. State of the queue at the blackout (08-29 23:25 cluster time)

Running: dv3 std s21/s22 (1M) + std-3M pack (s23-25, resume chain 3017793) + tg_clamp s2; r2d NOCLAMP dH/dDP s122/123; RLPD dHsucc_dup s30-32. Pending in order: 33 RLPD (dup, fails $\times 16$, dense $\times 12$), r2d w3 s84-87 $\times 2$ packs, DEMO3 smoke, then 35 DP replicates as filler from 08-31 00:23. Harvest files: `paper/harvest_2026-08-3{0,1}_*.md` on the cluster (+20/+44/+62 h). Post-blackout list: `paper/NEXT_2026-08-28.md` (bottom).

_version: 2026-08-30 01:55 (cluster clock) — FINAL pre-blackout stamp (dHsucc_dup s30-32 added). Numbers above are complete for: DP (n=10), RLPD (1a) sparse (n=8 v 8), r2dreamer corrected-world (n=4 v 4). Pending: RLPD dense/fails/dup, r2d n=8 and NOCLAMP finals, dv3 3M, DEMO3._